

Internship Report – Day 4

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Internship Program: ParkNSecure – Smart Parking Solutions

Day 4 Overview:

Today, I developed a Python script focuses on detecting and reading vehicle number plates from images using Python. OpenCV is used for image preprocessing and detecting the plate region through techniques like grayscale conversion, edge detection, and contour analysis. Once the number plate is located, EasyOCR extracts the alphanumeric text from the cropped region

Tools Used Today:

GEMINI, CHATGPT, VS CODER .

Task Given:

TO CREATE THE ANPR MODEL TO DETECT THE LETTERS AND NUMBERS PRESENT IN THE VECHILES NUMBERPLATES.

Approach to the Task:

The image is first preprocessed using OpenCV by converting it to grayscale, applying edge detection, and detecting contours. The region resembling a number plate is then extracted based on shape and size. This region is passed to EasyOCR to recognize and extract the text. The detected plate and text are finally displayed on the original image.

Attachments:-



authorized_list.csv > data	
1	KA01AB1234
2	AP16CD5678
3	TS09EF4321
4	NHZODV2366

```

main.py > ...
1  import cv2
2  import easyocr
3  import pandas as pd
4  import re
5  import os
6
7  def load_authorized_list(file_path="authorized_list.csv"):
8      if not os.path.exists(file_path):
9          print("⚠️ Authorized list file does not exist.")
10         return []
11         return pd.read_csv(file_path, header=None)[0].tolist()
12
13 def preprocess_image(image_path):
14     # Read the image
15     img = cv2.imread(image_path)
16     # Convert to grayscale
17     gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
18     # Apply thresholding to make the image binary (black and white)
19     _, thresh = cv2.threshold(gray, 150, 255, cv2.THRESH_BINARY_INV)
20     return thresh
21
22 def recognize_plate(image_path):
23     if not os.path.exists(image_path):
24         print(f"⚠️ The image file '{image_path}' does not exist.")
25         return None
26
27     # Preprocess the image
28     processed_image = preprocess_image(image_path)
29
30     reader = easyocr.Reader(['en'])
31     result = reader.readtext(processed_image)
32
33     for (bbox, text, prob) in result:
34         # Filter text length to detect license plate and clean it
35         if len(text) >= 7:
36             clean_text = clean_plate_text(text)
37             return clean_text
38
39     return None
40
41 def clean_plate_text(text):
42     # Remove non-alphanumeric characters and keep only the valid characters
43     cleaned_text = re.sub(r'[^A-Z0-9]', '', text.upper())
44     return cleaned_text
45
46 def check_access(plate_number, authorized_list):
47     return plate_number in authorized_list
48
49 if __name__ == "__main__":
50     # Specify the path to your image and authorized list CSV
51     image_path = "test_images/car1.jpg"
52     authorized_list_path = "authorized_list.csv"
53
54     # Load authorized plate numbers
55     authorized_plates = load_authorized_list(authorized_list_path)
56
57     if not authorized_plates:
58         print("⚠️ No authorized plates to check. Please check the authorized list.")
59     else:
60         plate = recognize_plate(image_path)
61
62         if plate:
63             print(f"Detected Plate: {plate}")
64             if check_access(plate, authorized_plates):
65                 print("✅ Access Granted. Opening Gate...")
66             else:
67                 print("❌ Access Denied.")
68         else:
69             print("⚠️ Plate not detected. Please check the image quality or angle.")
70

```

Detected Plate: RJILCV0002]

❌ Access Denied.

PS C:\Users\C H UDAY KUMAR\OneDrive\Desktop\parkNsecure\apnr> █

